

GIS AND THE ENVIRONMENT(77-701819)

HARDIKSINH AJAYSINH PARMAR

Table of Contents

1. Introduction.....	3
2. Study Area: Fanshawe Gate Farm	4
3. Methodology.....	6
3.1 Georeferencing and Digitizing.....	6
3.2 SRTM (Shuttle Radar Topography Mission) and Kriging interpolation	7
4. Result	10
4.1 Discussion of Result.....	14
5. Conclusion.....	14
References	16

List of Figures

Figure 1Fanshawe Gate Farm	5
Figure 2the Georeferenced Image	10
Figure 3Phase 1 habitat classification	11
Figure 4Topography	12
Figure 5Vegetation	13

1. Introduction

The Geographic Information System (GIS) is a tool for gathering, analysing, and displaying geographical information (Mierzejowska & Oga'a, 2018). Numerous ecosystems, species, and environmental variables have been investigated using this method. Habitat categorization, which entails the identification and mapping of distinct habitats based on their features, is one of the significant uses of GIS, as noted by Sousa, et al. (2016). This step is critical for environmental management and preservation as it forms the foundation for future conservation efforts and monitoring initiatives. (Sousa, et al., 2016).

In this report, we will be using GIS techniques to carry out a Phase 1 habitat categorization of a study area. The Phase 1 habitat categorization is a standard method used in the UK for mapping habitats. It involves the identification of broad habitat types based on a set of criteria, such as vegetation type, soil characteristics, and landform. This method has been widely adopted in the UK due to its simplicity and efficiency in mapping large areas. Phase 1 habitat categorization provides a useful framework for analysing the distribution of different habitats, and it can be easily integrated with other GIS techniques for environmental analysis. This method involves the collection of data on the physical, biological, and ecological features of a site to produce a map that illustrates the distribution of different ecosystems and significant species.

With respect to the study area, Fanshawe Gate Farm, the aim of this report is to evaluate the effectiveness of GIS techniques for environmental analysis. This includes the use of GIS to create detailed maps and images of the study area that can be used to analyse the ecological characteristics of the site.

To achieve this aim, the report has identified three specific objectives. The first objective is to produce a georeferenced image of the study area. This will involve collecting data on the location and boundaries of the site and using this information to create an accurate map that can be used as a reference point for further analysis.

The second objective is to produce a comprehensive map of the site that illustrates the distribution of different ecosystems and significant species. This will require the collection and analysis of data on the various habitats present in the study area, including wetlands, woodlands, grasslands, and agricultural fields. This map will help to identify areas of high biodiversity and ecological importance.

The third objective is to produce aspect, slope, and hillshade maps of the study area. These maps will provide detailed information on the topography of the site, including the orientation and angle of different slopes and the amount of shading that is present in different areas. This information can be used to identify areas that are vulnerable to erosion and other forms of environmental degradation, as well as to assess the suitability of different areas for different types of land use.

2. Study Area: Fanshawe Gate Farm

Fanshawe Gate Farm is a small, non-intensively managed livestock farm located in the southwest of Sheffield, UK, at 53°18'2" N latitude and 1°32'2" W longitude. The farm is part of the Environmental Stewardship scheme, which was established in 1992 and aims to promote sustainable farming practices. The scheme prohibits the use of fertilizers and excessive grassing in the area to help protect and enhance the natural environment.

The farm is divided into seven fields, with six of them coming under the Environmental Stewardship scheme. The majority of the fields are grasslands, while one of them is a grassland with shrubs. The northern field is known to support a diverse range of grassland fungi.

Hence, Fanshawe Gate Farm provides an excellent study area for environmental analysis using GIS techniques, as it is subject to specific regulations and is relatively small and well-defined. The Environmental Stewardship scheme provides a unique opportunity to assess the impact of such regulations on the natural environment and the effectiveness of GIS techniques in monitoring and analysing environmental changes.

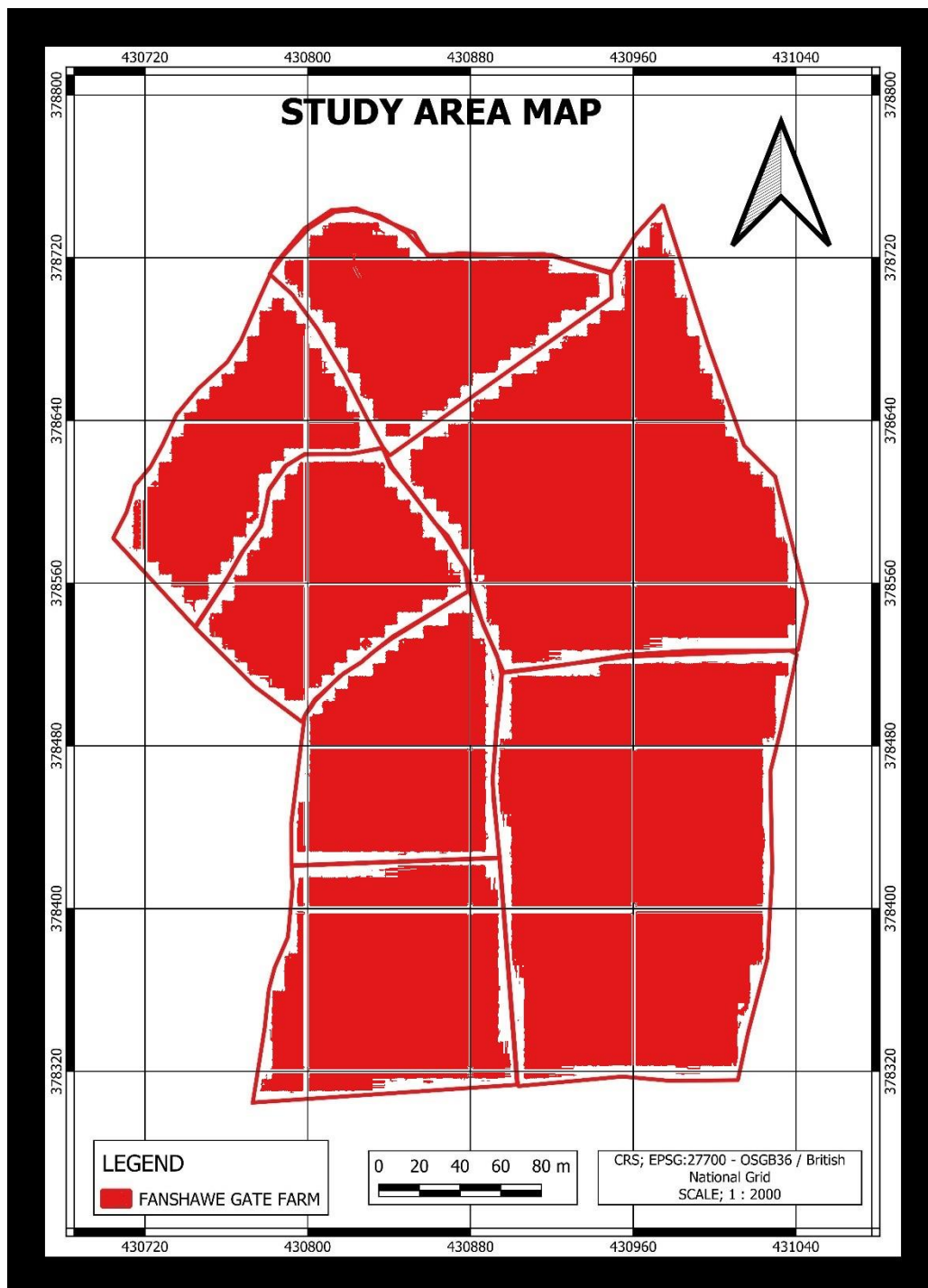


Figure 1 Fanshawe Gate Farm

3. Methodology

3.1 Georeferencing and Digitizing

Georeferencing is an essential technique used in GIS that plays a crucial role in assigning spatial coordinates to non-spatial data such as images or maps (Lahoti, et al., 2019). The process of georeferencing is critical for accurate mapping, analysis, and visualization of geographic data, as it enables the integration of disparate data sources into a common reference system. By associating spatial coordinates with non-spatial data, georeferencing facilitates the integration of diverse data types and helps in the creation of multi-layered maps that can be analyzed and visualized for a range of applications. The use of georeferencing in GIS enables spatial analysis to be conducted and helps in identifying patterns and relationships between different geographic features, which can provide insights into various environmental, social, and economic issues. Therefore, georeferencing is a fundamental process that serves as the foundation for many GIS applications and is a critical component in supporting informed decision-making processes across various fields (Lahoti, et al., 2019). The methodology of georeferencing used in this study involved the following steps:

1. Image selection: The initial step in the georeferencing process entails the careful selection of the image or map that necessitates spatial referencing. For the purposes of this particular study, a satellite image capturing Fanshawe Gate Farm was specifically chosen as the target for georeferencing to ensure accurate spatial alignment.
2. Choosing reference data: The subsequent step involves the selection of appropriate reference data, which typically consists of existing maps or ground control points that provide the necessary spatial coordinates for the non-spatial image. Within the context of this study, existing maps of the study area served as the chosen reference data for georeferencing purposes.
3. Adding control points: An integral aspect of the georeferencing process is the addition of control points to both the non-spatial image and the reference data. These control points serve as spatially corresponding locations that establish the linkage between the non-spatial image and the reference data. In this study, control points were meticulously added to both the satellite image and the existing maps, ensuring accurate alignment between the two.
4. Transformation: Following the establishment of control points, the subsequent step involves transforming the non-spatial image to achieve a harmonious match with the reference data. This transformation process relies on mathematical algorithms to

ensure accurate spatial referencing. For this study, the polynomial transformation method was employed to achieve the desired georeferencing outcome.

5. Checking accuracy: The final step of the georeferencing process involves a meticulous accuracy assessment of the georeferenced image. This is accomplished by comparing the georeferenced image with other spatial data and maps, effectively evaluating the alignment of features. In the context of this study, the accuracy of the georeferenced image was rigorously verified by overlaying it with existing maps of the study area, ultimately confirming the precise alignment of features and ensuring the reliability of the georeferencing process.

Georeferencing is a critical step in GIS analysis, and the methodology used in this study ensured that accurate spatial coordinates were assigned to the non-spatial satellite image of Fanshawe Gate Farm. This allowed for the production of detailed maps and analysis of environmental features in the study area(Bloom, et al., 2018).

In the case study of Fanshawe Gate Farm, digitizing was a key method used in the production of a comprehensive map of the site, which illustrated the distribution of different ecosystems and significant species. Digitizing is a process in GIS that involves converting hardcopy maps, aerial photographs, and other imagery into digital format(Kamel & Abu El Ella, 2016).

In this case, the digitizing process involved using a software program such as ArcGIS to trace the boundaries of different ecosystems and significant species on the hardcopy map of Fanshawe Gate Farm. The resulting digital data could then be used to create accurate and detailed maps of the site that could be used for further analysis(Manjula, et al., 2010).

Digitizing is an essential GIS job as it enables the creation of accurate and up-to-date maps that can be used for various applications such as land-use planning, resource management, and environmental analysis. The use of digitizing in the Fanshawe Gate Farm case study helped to produce a comprehensive map that provided valuable information on the distribution of different ecosystems and significant species, which can aid in the formulation of conservation plans and other environmental management strategies(Manjula, et al., 2010).

[3.2 SRTM \(Shuttle Radar Topography Mission\) and Kriging interpolation](#)

SRTM (Shuttle Radar Topography Mission) is an advanced remote sensing technique that harnesses the power of radar data to generate a highly accurate and detailed digital elevation model (DEM) of the Earth's surface. By employing a specially designed radar

system onboard the space shuttle, SRTM captures precise measurements of the topography, elevations, and surface features of our planet(Huggel, et al., 2008).

The data collected by SRTM allows for the creation of a comprehensive and comprehensive DEM that represents the varying elevations across vast areas of land. This invaluable resource enables scientists, researchers, and professionals in various fields to delve into the intricacies of terrain analysis, geomorphology, hydrology, and cartography, among others(Guth & Geoffroy, 2021).

One of the significant advantages of SRTM is its ability to penetrate cloud cover and capture accurate topographic information even in areas where traditional optical remote sensing methods would be hindered. This makes SRTM particularly valuable for regions with dense vegetation, rugged terrains, or frequent cloud cover(Lemenkova & Debeir, 2022).

The resulting DEM from SRTM serves as a powerful tool for a wide range of applications. For instance, in the field of terrain analysis, the SRTM DEM can be used to extract slope, aspect, and curvature information, enabling a comprehensive understanding of landforms and their characteristics. Additionally, floodplain mapping benefits from SRTM data by providing accurate elevation data to assess areas prone to flooding and develop effective flood mitigation strategies(Lemenkova & Debeir, 2022).

Furthermore, natural resource management greatly benefits from SRTM-derived elevation data. The DEM enables the identification of suitable areas for various land uses, such as agriculture, forestry, and urban planning. By analyzing the terrain characteristics derived from SRTM data, land managers can make informed decisions regarding resource allocation, conservation efforts, and sustainable development (Nwacholundu, et al., 2021).

In the case study of Fanshawe Gate Farm, SRTM data was used to create the aspect, slope, and hillshade maps of the study area. These maps provide important information about the topography of the land, which can be used to identify areas of high and low elevation, as well as areas that are steep or flat. This information is useful for a variety of applications, including land use planning, natural resource management, and environmental analysis(Lemenkova & Debeir, 2022).

The aspect map produced using SRTM data shows the orientation of the land surface with respect to the sun's rays. Areas facing the sun are shown in bright colours, while areas facing away from the sun are shown in darker colours. This information can be used to identify areas that are likely to have higher solar radiation, which may affect vegetation growth and other ecological processes(Lemenkova & Debeir, 2022).

The slope map produced using SRTM data shows the angle of the land surface with respect to the horizontal plane. Areas with steep slopes are shown in bright colours, while areas with gentle slopes are shown in darker colours. This information can be used to identify areas that are prone to erosion or landslides, as well as areas that may be difficult to access for agricultural or other land uses.

Finally, the hillshade map produced using SRTM data shows the three-dimensional shape of the land surface, with shadows and highlights added to simulate the effect of sunlight. This map can be used to identify areas of relief, such as ridges and valleys, as well as areas of uniform slope or elevation. This information can be used for a variety of applications, including visualizing the landscape and identifying areas of potential ecological importance (Nwacholundu, et al., 2021).

In the case study, Kriging interpolation was used to create aspect, slope, and hillshade maps of the study area. Kriging is a geostatistical interpolation method that predicts unknown values of a variable at unmeasured locations based on measurements taken at known locations. It is commonly used in GIS applications for spatial analysis and modelling (Androo, et al., 2015).

Kriging works by estimating the spatial autocorrelation of the variable being analysed and then using this information to generate a weighted average of the surrounding measured values. This weighted average is then used to estimate the value of the variable at the unmeasured location. The resulting interpolated surface is generally smoother than the original data and can be used for further analysis or visualization (Pereira, et al., 2022).

In the case study, Kriging was used to interpolate elevation data and create an aspect, slope, and hillshade maps. These maps were used to better understand the topography of the study area and to inform habitat categorization. Kriging interpolation is a powerful tool in GIS analysis and can be used in a wide range of applications, including environmental modeling, urban planning, and natural resource management (Pereira, et al., 2022).

4.Result

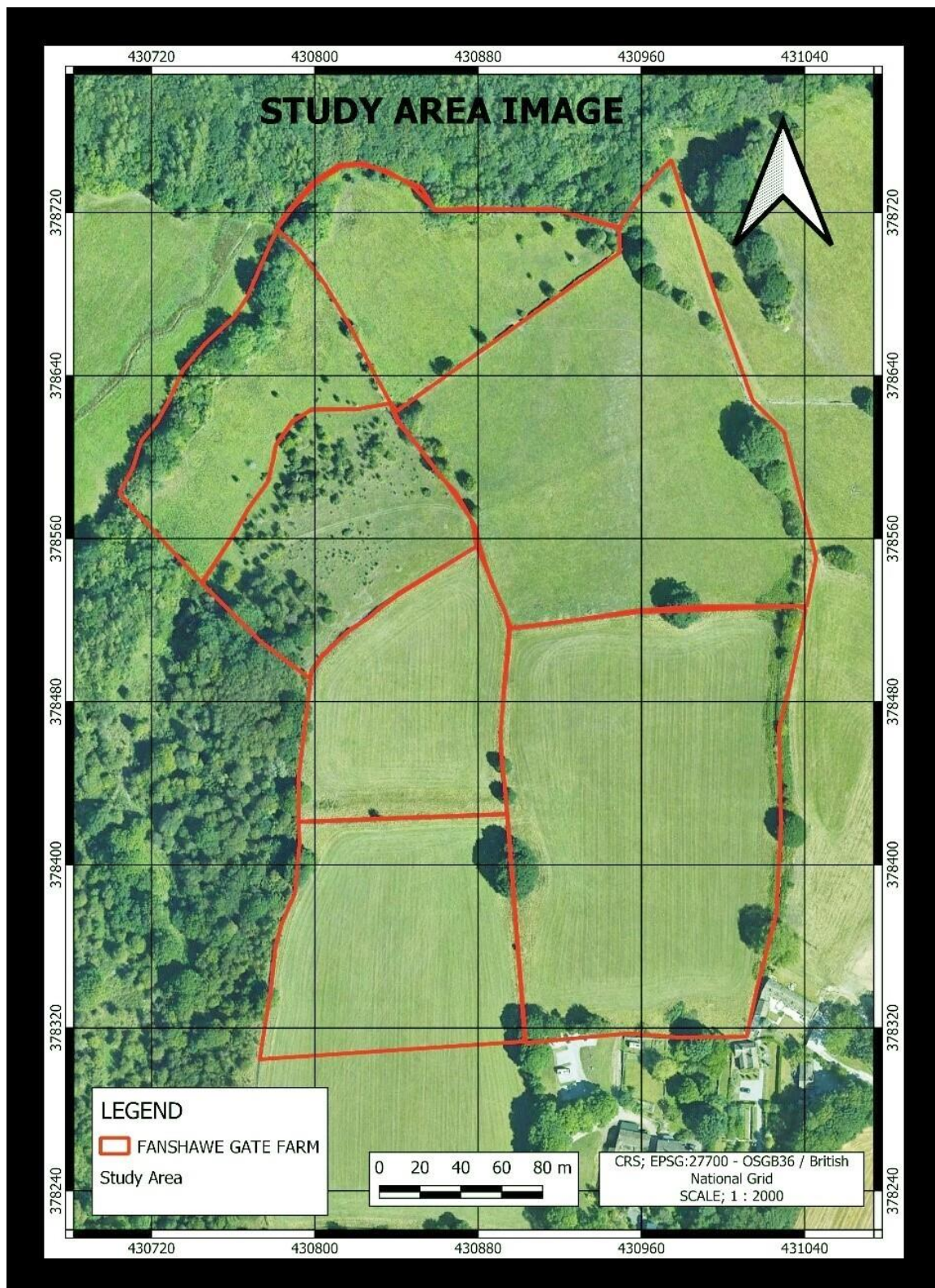
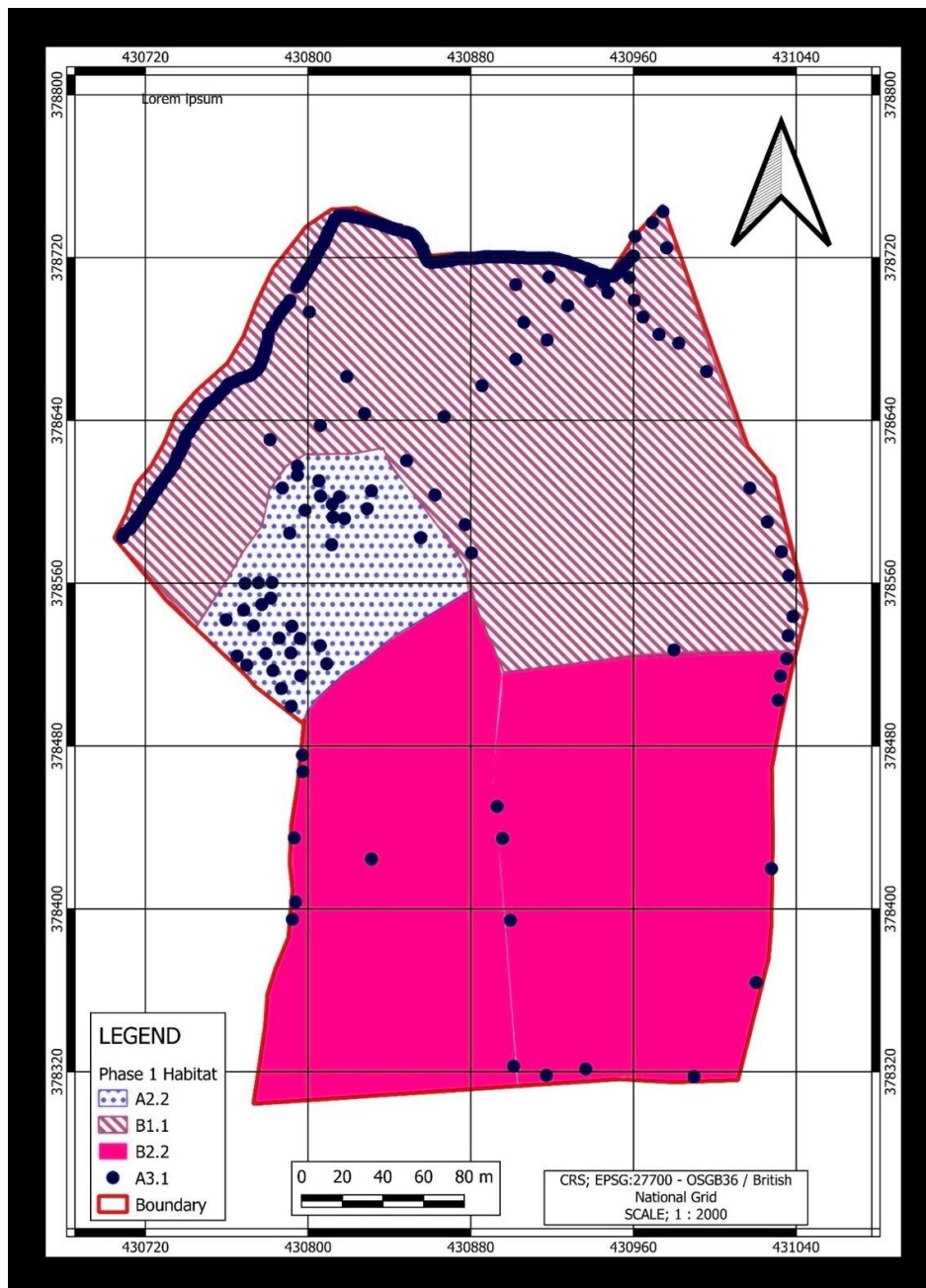


Figure 2the Georeferenced Image

The georeferenced image produced for Fanshawe Gate Farm in Figure 2 provided a spatially accurate representation of the study area, enabling accurate analysis and

interpretation of environmental data. It allowed for the identification and mapping of different habitats, including acid grassland, neutral grassland, scrub, and broadleaved parkland. This information is essential for land management and conservation planning, as it allows for targeted efforts to be made to protect and enhance important habitats and species. Overall, the georeferenced image proved to be a valuable tool for environmental analysis in the Fanshawe Gate Farm case study.



The map above, Figure 3, revealed that Fanshawe Gate Farm is categorized into four groups,

including scattered scrub (A.2.2), unimproved acid grassland (B.1.1), semi-improved neutral grassland (B.2.2), and broadleaved parkland with scattered trees (A.3.1).

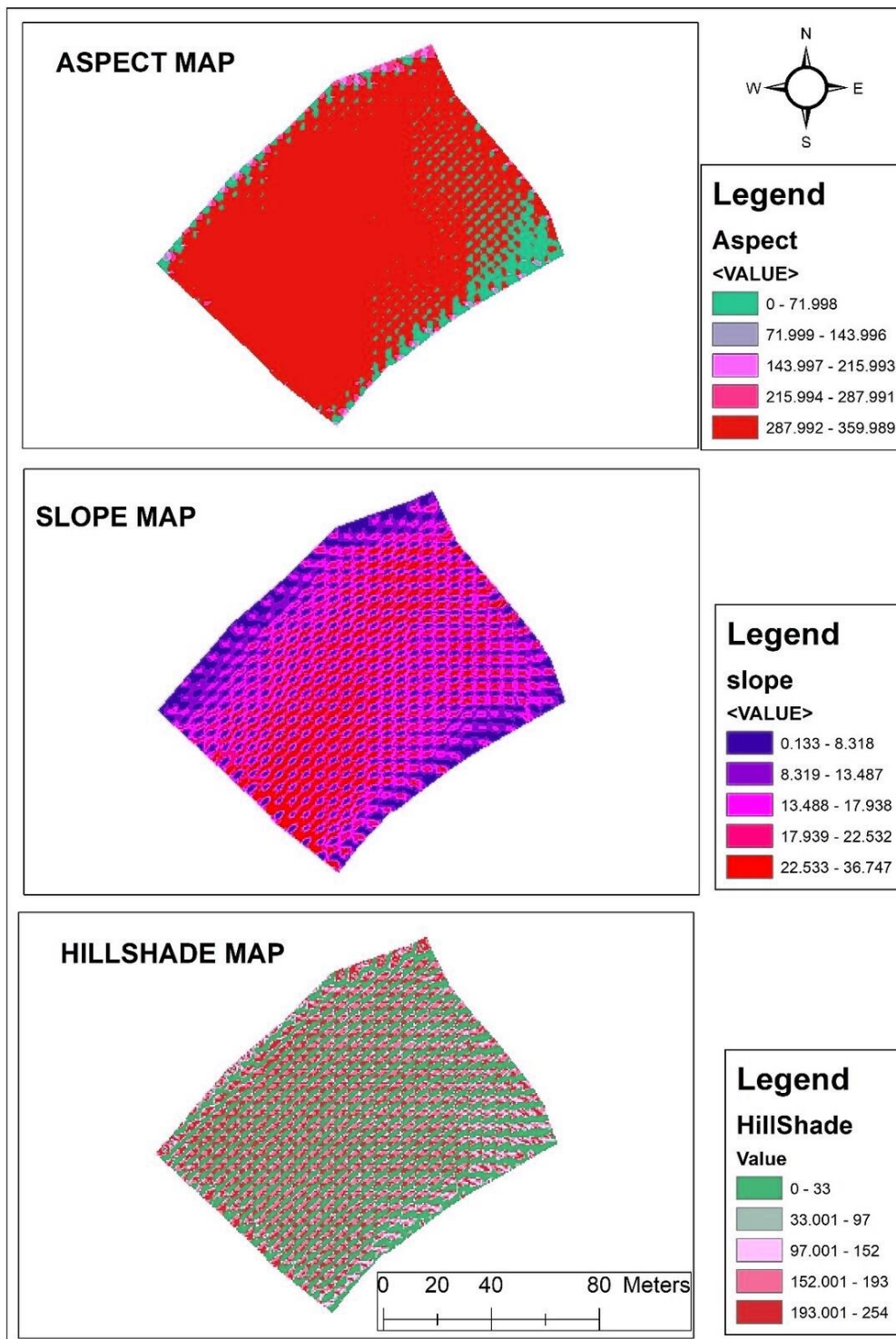


Figure 4Topography

Figure 4 is the topography map produced using GIS techniques. The topography map provides a visual representation of the land surface's physical features, such as elevation, slope, and aspect. The map was created using SRTM data, which provided elevation data for the study area. The resulting map showed a range of elevations across the farm, with the

highest point in the southern part of the farm and the lowest point in the northeastern corner. The slope map and aspect map were also created from the same SRTM data. The slope map showed the steepness of the land surface across the farm, while the aspect map showed the direction that the land surface faced.

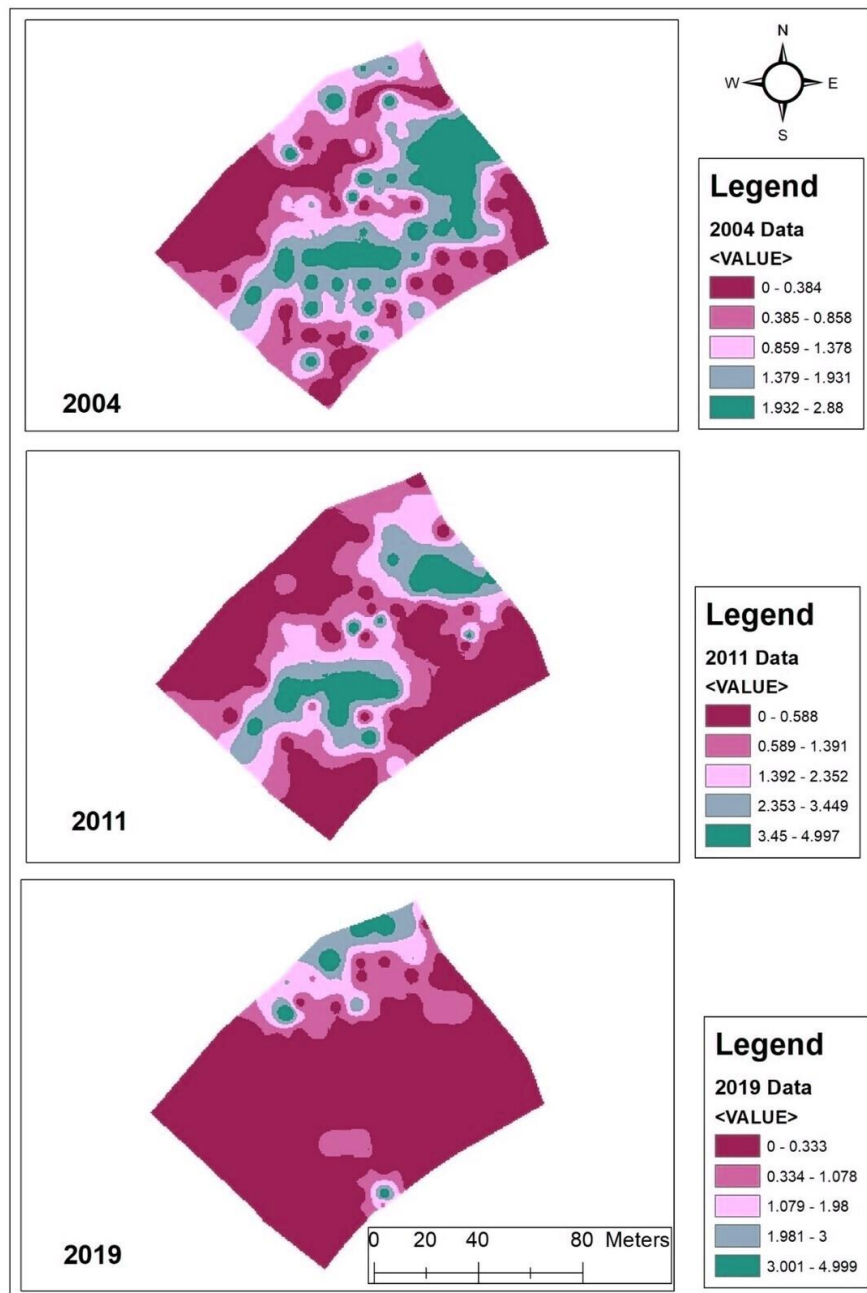


Figure 5Vegetation

For the Fanshawe Gate Farm case study, Figure 5 is a vegetation map produced using GIS techniques. The map illustrates the distribution of different ecosystems and significant species in the study area. The vegetation map was created through the use of remote sensing data, field surveys, and GIS analysis.

4.1 Discussion of Result

The results of the Fanshawe Gate Farm case study provide valuable insights into the land cover and habitat classification of the study area. The analysis revealed that the farm is classified into four main habitat types: scrub-scattered, acid grassland-unimproved, neutral grassland-semi-improved, and broadleaved parkland/scattered trees.

The scrub-scattered habitat type, which is classified as A.2.2, covers a small portion of the study area. This habitat type is characterized by areas of dense scrub with scattered trees and shrubs, and it provides habitat for a variety of bird species.

The acid grassland-unimproved habitat type, classified as B.1.1, covers the largest area of the study site. This habitat type is characterized by the presence of heather and other acid-loving plants, and it supports a diverse range of invertebrates and birds.

The neutral grassland-semi-improved habitat type, classified as B.2.2, is characterized by a mix of grasses and wildflowers, and it provides important habitat for a range of bird and insect species. This habitat type covers a relatively small portion of the study site.

Finally, the broadleaved parkland/scattered trees habitat type, classified as A.3.1, covers a very small portion of the study site. This habitat type is characterized by scattered trees and shrubs in a grassland setting, and it provides important habitat for a range of bird species.

Hence, the habitat classification map produced in this study provides a valuable tool for land managers and conservationists to better understand the distribution of different habitats on the Fanshawe Gate Farm, and to develop management strategies that will help protect and enhance these important habitats.

5. Conclusion

This comprehensive case study has unequivocally showcased the remarkable effectiveness and profound impact of Geographic Information System (GIS) techniques in facilitating in-depth environmental analysis. By employing a holistic approach that encompassed georeferencing, digitizing, SRTM, and Kriging interpolation, a meticulous and highly

informative map of the renowned Fanshawe Gate Farm has been meticulously crafted, delving into the intricate details of its diverse habitats and species.

Georeferencing, a fundamental process in GIS, played a pivotal role in accurately assigning spatial coordinates to the non-spatial image, ensuring the precise mapping and analysis of the geographic data. Through the selection of the most suitable satellite image of Fanshawe Gate Farm, georeferencing provided a solid foundation for subsequent analyses and interpretations.

The digitizing process further enriched the GIS workflow by converting valuable analog information into digital format. By meticulously capturing and converting existing maps and ground control points into digital representations, the digitizing step solidified the accuracy and reliability of the spatial data, ensuring seamless integration with the broader GIS framework.

Incorporating Shuttle Radar Topography Mission (SRTM) data, an advanced remote sensing technique, added another layer of depth and precision to the environmental analysis. By harnessing the power of radar data, a highly detailed digital elevation model (DEM) of the farm's terrain was derived, unveiling intricate topographic characteristics crucial for assessing landforms, floodplain mapping, and natural resource management.

Furthermore, the application of Kriging interpolation, a sophisticated geostatistical technique, enhanced the precision and reliability of the obtained results. By intelligently inferring and interpolating values at unmeasured locations based on nearby sampled data points, Kriging interpolation facilitated the creation of accurate maps depicting the distribution of different ecosystems and significant species across the farm.

The culmination of these GIS techniques resulted in a comprehensive and visually captivating map, effectively classifying Fanshawe Gate Farm into four distinct habitat groups: scrub, acid grassland, neutral grassland, and broadleaved parkland/scattered trees. Such delineation provides invaluable insights into the farm's current ecological state, acting as a crucial foundation for informed decision-making and future conservation efforts.

Undoubtedly, the outcomes of this case study shed light on the immense potential of GIS techniques in environmental analysis. The meticulous mapping and classification of habitats through the integration of georeferencing, digitizing, SRTM, and Kriging interpolation have not only enhanced our understanding of Fanshawe Gate Farm's ecological composition but

have also paved the way for informed conservation strategies that ensure the preservation and sustainable management of its diverse ecosystems.

References

- Androo, P. G., Herbei, M. & Herbei, R., 2015. Using the Arcgis spatial and 3d analysis tools for a sustainable development of localities. *Research Journal of Agricultural Science*, Volume 47, p. 4.
- Bloom, T. D., Flower, A. & DeChaine, E. G., 2018. Why georeferencing matters: Introducing a practical protocol to prepare species occurrence records for spatial analysis. *Ecology and evolution*, 8(1), pp. 765-777.
- Guth, P. L. & Geoffroy, T. M., 2021. LiDAR point cloud and ICESat-2 evaluation of 1 second global digital elevation models: Copernicus wins. *Transactions in GIS*, 25(5), pp. 2245-2261.
- Huggel, C. et al., 2008. Evaluation of ASTER and SRTM DEM data for lahar modeling: a case study on lahars from Popocatépetl Volcano, Mexico. *Journal of Volcanology and Geothermal Research*, 170(1-2), pp. 99-110.

- Kamel, M. & Abu El Ella, E. S., 2016. Integration of Remote Sensing & GIS to Manage the Sustainable Development in the Nile Valley desert fringes of Assiut-sohag governorates, Upper Egypt. *Journal of the Indian Society of Remote Sensing*, Volume 44, pp. 759-774.
- Lahoti, S., Kefi, M., Lahoti, A. & Saito, O., 2019. Mapping methodology of public urban green spaces using GIS: An example of Nagpur City, India. *Sustainability*, 11(7), pp. 21-66.
- Lemenkova, P. & Debeir, O., 2022. Satellite Image Processing by Python and R Using Landsat 9 OLI/TIRS and SRTM DEM Data on Côte d'Ivoire, West Africa. *Journal of imaging*, 8(12), p. 317.
- Manjula, K. R., Jyothi, S. & Varma, S. A., 2010. Digitizing the forest resource map using ArcGIS. *International Journal of Computer Science Issues (IJCSI)*, 7(6), p. 300.
- Mierzejowska, A. & Żogała, M., 2018. The characteristics of geographical information systems in terms of their current use. *Journal of water and land development*, Volume 39, pp. 101-108.
- Nwacholundu, U. V.; Izuchukwu, I. J.; Ebele, E. J.; Onyedika, E. J.; & Chinagorom, I. E., 2021. Generating and analyzing terrain characteristics from shuttle radar topographic mission (SRTM), DEM. *World Journal of Advanced Research and Reviews*, 10(3), pp. 198-206.
- Pereira, G. W.; Valente, D. S.M.; de Queiroz, D. M.; Santos, N. T.; & Fernandes-Filho, E. I., 2022. Soil mapping for precision agriculture using support vector machines combined with inverse distance weighting. *Precision Agriculture*, 23(4), pp. 1189-1204.
- Sousa, L. P., Sousa, A. I., Alves, F. L. & Lillebø, A. I., 2016. Ecosystem services provided by a complex coastal region: challenges of classification and mapping. *Scientific Reports*, 6(1), pp. 1-14.